

STAGE 1: Raw Data Selection



SATD Repayment Dataset



- Language: Python & Java
- Original Curation: ≥ 3 -word SATD
- Instances: $>156K$

Candidate Repository Selection

- Stars: > 500
- Curation: Diverse, Realistic system

3,968 Candidates

Function-level Raw Data

STAGE 2: Design Issue Curation

LLM Judge Filtering



Comment



Pre/Post code



Commit messages

- ✓ Design issue?
- ✓ Pure design?
- ✓ Semantic difference?
- ✓ Issue addressed?

4 positive criteria & Confidence ≥ 6

447 Candidates

Manual Verification & Adjudication



Comment



Pre/Post code



Commit messages



LLM decision & rationale

- 4-5 positive votes: **ACCEPTED**
- 3 positive votes: **ADJUDICATION**
- 2 or fewer votes: **REJECTED**

STAGE 3: Repository-Level Instance Construction

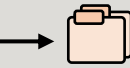
LLM-Based Data Pipeline

1

1 Patch Isolation



Developer patch



Candidate patch

Remove unrelated edits (tests, docs)



3

3 Test Reuse & Construction

Comment

Repo snapshot

Dev./Can. patches

Search & reuse Existing tests

Reusable trigger/regression

Constructed trigger/regression



Test suites & FAIL/PASS2PASS

215 Candidates

Repository-level Candidates

Final Manual Artifact Verification & Adjudication

1

1 Gold Patch Verification

- Inspect patches & execution evidence
- Check relevance, correctness & refine

2

2 Weak Trigger Adjudication

- Incomplete patch $\rightarrow$ revise gold patch
- Invalid test $\rightarrow$ rewrite trigger test

3

3 Test Suites Verification

- Score test quality from 0 to 3
- ≥ 1 low-quality: discuss & rewrite

215 Repository-Level Instance

Docker-based re-execution loop